STAT 8240 DATA MINING II

Class Project
Kennesaw State University
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Project Context & Motivation

- Focus: Moving beyond static, unimodal, tabular data
- Project: Longitudinal & Multimodal Deep Learning
- Challenges in high-stakes fields (healthcare, finance, climate science):
  - **Longitudinal:** multiple observations per entity over time
  - **Irregular:** varying sampling rates
  - **Asynchronous:** modalities start/stop at different times
  - **Multimodal:** combining data from multiple sources (e.g., sensor + text + imaging)
- Goal: Build deep learning architectures that can handle complex temporal dynamics and multimodal interactions

Anchor Paper: TIME-IMM (2025)

- Benchmark for irregular, multimodal, multivariate time series
- Key features:
  - 9 example datasets for forecasting (e.g., MIMIC for healthcare, FNSPID for finance)
  - IMM-TSF Library for forecasting on irregular multimodal data
- Focus: Provide a starting point for reproducibility, application, or methodological extension

Project Tracks

- Track A – Reproducibility: Replicate results from one or more datasets using IMM-TSF
- Track B – Application: Apply IMM-TSF to a dataset of your choice
- Track C – Methodological Extension: Modify architecture (e.g., new modality fusion module)

Grading

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Final Project: 30%

- Proposal & Progress Check (5%)
- Final Report (15%)
- Final Submission Code: (5%)
- Final Presentation: (5%)
- (Optional) Analytics Day Bonus (5%)

Proposal & Weekly Progress Check (5%)

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- Goal: Keep students on track
- Weekly 3-5 minutes presentation with 1-2 slides per groups
- Beginning of Thursday Class (April 2nd , April 9th , April 16th)
- What I expect:
  - April 2nd (short proposal): Problem formulation, data introduction, methods
  - April 9th : Report progress, initial results
  - April 16th : Report progress, major results
  - (optional) Submit poster to Analytics Day by April 20th

Final Class Presentation (5%)

- Date: April 23rd or May 4th (depending on progress)
- ~15 minutes per group
- Focus:
  - Problem & data description
  - Methodology
  - Results & insights

Final Project Report (15%)

- 6–10 pages recommended
- Suggested structure
  - Problem formulation
  - Data description
  - Methodology
  - Results & evaluation
  - Discussion & limitations
- Due: May 4th

Code (5%)

- Collaborative GitHub repository
- Tracking via commit logs
- Requirements:
  - Clean and well-structured code
  - Proper documentation (README)
  - Reproducibility (instructions, seeds)
- Due: May 4th

Analytics Day (Extra Credit: +5%)

- Optional external presentation
- Opportunity to showcase work
- Poster submission due: April 20th
- Bonus
  - +5% added to final grade

Team

- Teams consist of 1-3 members
- Collaborative code repository
  - Tracking via commit logs
- Implementation via python
- Each team member must understand the methodology
- You are encouraged to use AI tool! But each team member must understand the methodology! I will ask questions during your presentation!

Questions?